

STAT 332 Sampling and Experimental Design

University of Waterloo - Fall 2025

19th September 2025

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1 Sampling

Lecture 1, 2025/09/03

1.1 Intro & Survey Terminology

Assignments are there to help you prepare for exams!

Why study survey sampling?

- Obtain data for a question of interest from a population.
- Cannot take a census.
- Finite time and resources.

Definition 1.1 (Target Population).

The complete collection of individuals we wish to study and draw conclusions about. Denoted by $U = \{1, 2, \dots, N\}$.

Definition 1.2 (Sampled (Study) Population).

The collection of individuals that could be included by our sampling design.

Definition 1.3 (Sampling Unit).

The object being sampled.

Definition 1.4 (Frame).

The list of all sampling units.

Definition 1.5 (Observation Unit).

The object on which we take measurements.

Note. Often, this is the person we ask questions to.

Three main steps at which error and bias can be introduced in the conducting of a survey:

- Constructing the frame (coverage error or study error)

- Obtaining the sample/respondents (sampling bias/error, non-response error)
- Measurement of the response (measurement error)

Definition 1.6 (Sampling Design (Sampling Protocol)).

A **sampling design** (or sampling protocol) is the mechanism by which we select our samples.

Note. **Sampling error** is the estimation error due to sampling variability.

Definition 1.7 (Selection Bias).

Differences between the target population and the effective sampled population.

Note. Ideally, target population = population that can be effectively sampled.

Sources of selection bias:

- Coverage: e.g., can be difficult to construct a complete frame
- Sampling: e.g., improper sampling design
- Non-response: people don't respond or are not reachable

Definition 1.8 (Measurement Error).

The measured values of the attributes on the observation units are different from the true values.

Lecture 2, 2025/09/08

1.2 Probability-Based Sampling Design

Assume that frame is complete (ideal case). Assume also that $U = \{1, 2, \dots, N\}$. We denote the responses (corresponding values for attribute of interest) by $y_1, y_2, \dots, y_N$.

Population quantities:

- $\mu = \frac{1}{N} \sum_{i \in U} y_i$ (population mean)
- $\tau = \sum_{i \in U} y_i$ (population total)
- $\sigma^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 = \frac{1}{N-1} [\sum_{i \in U} y_i^2 - N\mu^2]$ (population variance b/c population is finite).

Note.

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 \\
&= \frac{1}{N-1} \sum_{i \in U} (y_i^2 - 2\mu y_i + \mu^2) \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - 2\mu \sum_{i \in U} y_i + N\mu^2 \right] \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - 2N\mu^2 + N\mu^2 \right] \\
&= \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - N\mu^2 \right].
\end{aligned}$$

Suppose we have a sample $s \subseteq U$, a subset of frame with sample size n .

Estimates of population quantities based on sample:

- $\hat{\mu} = \frac{1}{n} \sum_{i \in s} y_i$, sample mean estimates population mean
- $\hat{\tau} = \sum_{i \in s} y_i$, sample total estimates population total
- $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i \in s} (y_i - \hat{\mu})^2$, sample variance estimates population variance

Note. As in STAT 231, we treat estimates as constant values based on a specific sample, i.e. y_i 's are constants. We treat the sampling design/process as random, so s is a realization of a random variable S .

Sampling design tells us $\mathbb{P}(S = s)$ which captures the **inclusion probabilities**.

Definition 1.9 (Inclusion Probabilities).

$$\pi_i = \mathbb{P}(i \in S), \quad \pi_{ij} = \mathbb{P}(i, j \in S), \quad i \neq j \in U.$$

Then, we can define estimators as functions of S :

- $\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i$ (sample mean estimator)
- $\tilde{\tau} = \sum_{i \in S} y_i$ (sample total estimator)
- $\tilde{\sigma}^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \tilde{\mu})^2$ (sample variance estimator)

Definition 1.10 (Simple Random Sampling (SRS)).

SRS is also known as **SRS without replacement (SRSWOR)**. Take a sample of size n out of N , such that all possible samples of size n are equally likely.

Note. The randomness in SRS comes from the sampling design, not the responses.

Example. Let $N = 4$, $n = 2$, then there are $\binom{4}{2} = 6$ possible samples:

$$\{1, 2\} \quad \{1, 3\} \quad \{1, 4\} \quad \{2, 3\} \quad \{2, 4\} \quad \{3, 4\},$$

each with probability $\frac{1}{6}$, e.g. $\mathbb{P}(S = \{1, 2\}) = \frac{1}{6}$.

Question: How many possible samples of size n ?

Answer: $\binom{N}{n} = \frac{N!}{n!(N-n)!}$.

Thus, we have

$$\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}} \quad \text{for any subset } s \text{ of size } n.$$

We can also compute inclusion probabilities:

$$\pi_i = \mathbb{P}(i \in S) = \frac{\# \text{ of samples with unit } i \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{1}{1}\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{n}{N}.$$

Note. $\binom{1}{1}$ is the number of ways to choose unit i (to include i , we must pick i , so 1 way), and $\binom{N-1}{n-1}$ is the number of ways to choose the remaining $n - 1$ units from the other $N - 1$ units.

$$\pi_{ij} = \mathbb{P}(i, j \in S) = \frac{\# \text{ of samples with units } i, j \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{2}{2}\binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}.$$

Note. $\pi_{ij} \neq \pi_i\pi_j$ because sampling from a finite population without replacement means that the events $i \in S$ and $j \in S$ are not independent.

R Intro & SRS in R

Here is some basic syntax for R:

```
x <- c()      # creates an empty vector
for (i in 1:50) {
  x[i] <- 2*i    # fill it with some stuff
}
x             # look at what it now has
mean(x)
var(x)        # sample mean and variance
```

Obtaining SRS in R:

```
> N <- 10
> n <- 5
> x = c(1:10)
> sample(x,5, replace=F)    # replace=F indicates without replacement
[1] 6 2 4 7 8
> sample(x,5, replace=F)
[1] 8 4 5 6 3
```

Probability Review

Let X be a discrete random variable. Then,

$$(1) \mathbb{E}[X] = \sum_x x \mathbb{P}(X = x), \quad \mathbb{E}[g(X)] = \sum_x g(x) \mathbb{P}(X = x).$$

$$(2) \text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

$$(3) \text{Linearity: } \mathbb{E}\left[\sum_i^n a_i X_i\right] = \sum_i^n a_i \mathbb{E}[X_i].$$

$$(4) \text{Var}\left(\sum_i^n a_i X_i\right) = \sum_i^n a_i^2 \text{Var}(X_i) + 2 \sum_i^n \sum_{j \neq i}^n a_i a_j \text{Cov}(X_i, X_j).$$

$$(5) \text{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i] \mathbb{E}[X_j].$$

$$(6) \text{ If } X_i \text{ and } X_j \text{ are independent, then } \text{Cov}(X_i, X_j) = 0.$$

Example. Let's define an indicator random variable. Let

$$I_i = \begin{cases} 1, & \text{if unit } i \text{ is selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i = 1) = \mathbb{P}(i \in S) = \pi_i$ and $\mathbb{P}(I_i = 0) = 1 - \pi_i$. Thus,

$$\mathbb{E}[I_i] = 0 \cdot (1 - \pi_i) + 1 \cdot \pi_i = \pi_i,$$

$$\mathbb{E}[I_i^2] = 0^2 \cdot (1 - \pi_i) + 1^2 \cdot \pi_i = \pi_i,$$

$$\text{Var}(I_i) = \mathbb{E}[I_i^2] - (\mathbb{E}[I_i])^2 = \pi_i - \pi_i^2 = \pi_i(1 - \pi_i).$$

For $i \neq j$, consider

$$I_i I_j = \begin{cases} 1, & \text{if both } i \text{ and } j \text{ are selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i I_j = 1) = \mathbb{P}(i, j \in S) = \pi_{ij}$ and $\mathbb{P}(I_i I_j = 0) = 1 - \pi_{ij}$. Thus,

$$\mathbb{E}[I_i I_j] = 0 \cdot (1 - \pi_{ij}) + 1 \cdot \pi_{ij} = \pi_{ij},$$

$$\text{Cov}(I_i, I_j) = \mathbb{E}[I_i I_j] - \mathbb{E}[I_i] \mathbb{E}[I_j] = \pi_{ij} - \pi_i \pi_j.$$

So, we can write $\tilde{\mu}$ in terms of indicators.

$$\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i = \frac{1}{n} \sum_{i=1}^N I_i y_i = \frac{1}{n} \sum_{i \in U} I_i y_i = \frac{1}{n} \sum_{i=1}^N I_i y_i.$$

Definition 1.11 (Bias, Unbiased Estimator).

Suppose that $\tilde{\theta}$ is an estimator for θ . Then, the **bias** of $\tilde{\theta}$ is

$$\text{Bias}(\tilde{\theta}) = \mathbb{E}[\tilde{\theta}] - \theta.$$

$\tilde{\theta}$ is an **unbiased estimator** for θ if $\text{Bias}(\tilde{\theta}) = 0$, i.e. $\mathbb{E}[\tilde{\theta}] = \theta$.

Question: What are $\mathbb{E}[\tilde{\mu}]$ and $\text{Var}(\tilde{\mu})$ for SRS?

Answer:

$$\begin{aligned}\mathbb{E}[\tilde{\mu}] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^N I_i y_i\right] = \frac{1}{n} \sum_{i=1}^N y_i \underbrace{\mathbb{E}[I_i]}_{\pi_i} \\ &= \frac{1}{n} \sum_{i=1}^N y_i \frac{n}{N} = \frac{1}{N} \sum_{i=1}^N y_i = \mu.\end{aligned}$$

So $\tilde{\mu}$ is an unbiased estimator for μ .

For SRS,

$$\text{Cov}(I_i, I_j) = \pi_{ij} - \pi_i \pi_j = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Then,

$$\begin{aligned}\text{Var}(\tilde{\mu}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^N I_i y_i\right) \\ &= \frac{1}{n^2} \left[\sum_{i=1}^N y_i^2 \underbrace{\text{Var}(I_i)}_{\pi_i(1-\pi_i)} + 2 \sum_{i=1}^N \sum_{j \neq i} y_i y_j \text{Cov}(I_i, I_j) \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \left[\sum_{i=1}^N y_i^2 - \frac{1}{N-1} \sum_{i=1}^N \sum_{j \neq i} y_i y_j \right]\end{aligned}$$

Note that $\sum_{i=1}^N \sum_{j \neq i} y_i y_j = \sum_{i=1}^N y_i \sum_{j \neq i} y_j = \sum_{i=1}^N y_i (N\mu - y_i) = N\mu \sum_{i=1}^N y_i - \sum_{i=1}^N y_i^2 = N^2\mu^2 - \sum_{i=1}^N y_i^2.$

$$\begin{aligned}&= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \left[\sum_{i=1}^N y_i^2 - \frac{1}{N-1} (N^2\mu^2 - \sum_{i=1}^N y_i^2) \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \frac{1}{N-1} \left[((N-1) + 1) \sum_{i=1}^N y_i^2 - N^2\mu^2 \right] \\ &= \frac{1}{n^2} \left(1 - \frac{n}{N}\right) \frac{n}{N} \frac{1}{N-1} \left(\sum_{i=1}^N y_i^2 - N\mu^2 \right) \\ &= \left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n}.\end{aligned}$$

Thus for SRS, $\mathbb{E}[\tilde{\mu}] = \mu$ and $\text{Var}(\tilde{\mu}) = \left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n}$, where $\left(1 - \frac{n}{N}\right)$ is called the **finite population correction (fpc)**.

To get CIs, we estimate $\text{Var}(\tilde{\mu})$ with $\widehat{\text{Var}}(\tilde{\mu}) = \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}$, as it turns out that $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.

Lecture 4, 2025/09/15

Exercise: Show that $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$ (PS2).

In practice, we need to estimate $\text{Var}(\tilde{\mu})$ because we don't know σ^2 .

Question: What is an unbiased estimator of $\text{Var}(\tilde{\mu})$?

Answer: Note that $\mathbb{E}\left[\left(1 - \frac{n}{N}\right) \frac{\tilde{\sigma}^2}{n}\right] = \left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n}$ (hint for A1 Q3).

Suppose we now have data from SRS, and we want an estimate of the population mean and 95% CI.

We care about CIs because we want to know how uncertain our estimate is.

Definition 1.12 (Standard Error (SE)).

Standard error (SE) is an estimate of the standard deviation (of an estimator) based on a specific sample.

Example. For μ based on a SRS, the estimate is $\hat{\mu}$. We estimate $\text{Var}(\tilde{\mu})$ with

$$\widehat{\text{Var}}(\tilde{\mu}) = \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n},$$

and so $\text{SE}(\hat{\mu}) = \sqrt{\widehat{\text{Var}}(\tilde{\mu})} = \sqrt{\left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}}$, note that “hat” is used for σ^2 .

For sampling, we will use Normal approximation to construct CIS, e.g. 95% CI for μ is

$$\hat{\mu} \pm 1.96 \text{SE}(\hat{\mu}).$$

Example. For τ (population total) in SRS, since $\tau = \sum_{i \in U} y_i = N\mu$, we have $\hat{\tau} = N\hat{\mu}$,

$$\text{Var}(\tilde{\tau}) = \text{Var}(N\tilde{\mu}) = N^2 \text{Var}(\tilde{\mu}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n},$$

so $\widehat{\text{Var}}(\tilde{\tau}) = N^2 \widehat{\text{Var}}(\tilde{\mu})$ and $\text{SE}(\hat{\tau}) = N \cdot \text{SE}(\hat{\mu})$. An approximate 95% CI for τ is

$$N\hat{\mu} \pm 1.96N \cdot \text{SE}(\hat{\mu}).$$

Example (Special Case).

For proportion, suppose response is binary i.e.

$$y_i = \begin{cases} 1, & \text{answered yes,} \\ 0, & \text{answered no.} \end{cases}$$

The population mean of y_i 's is called a **proportion**. Denote the population proportion as p . In this case,

$$\begin{aligned} \sigma^2 &= \frac{1}{N-1} \left(\sum_{i \in U} y_i^2 - N\mu^2 \right) \\ &= \frac{1}{N-1} \left(\sum_{i \in U} Np - Np^2 \right) \quad (\text{since } y_i^2 = y_i \text{ and } \mu = p) \\ &= \frac{N}{N-1} p(1-p). \end{aligned}$$

Based on SRS,

$$\hat{\sigma}^2 = \frac{n}{n-1} \hat{p}(1-\hat{p}),$$

where $\hat{p}$ is the sample proportion. So a 95% CI for p is

$$\hat{p} \pm 1.96 \text{SE}(\hat{p}) = \hat{p} \pm 1.96 \sqrt{\left(1 - \frac{n}{N}\right) \frac{\hat{p}(1-\hat{p})}{n-1}}.$$

Note that $\frac{n}{N}$ is the **sampling fraction** and $1 - \frac{n}{N}$ is the **fpc**. So if $n \approx N$, then $\text{Var}(\hat{\mu}) \rightarrow 0$, i.e. variability due to sampling $\rightarrow 0$. If $N \gg n$, then $\text{fpc} = 1 - \frac{n}{N} \approx 1$.

Recall: How many Americans ($N = 342M$) vs. Canadians ($N = 41.5M$) for a sample? If $\hat{\sigma}^2$ is similar for US/Canada, then $\text{SE}(\hat{\mu})$ is also very similar, even though N is much larger for US.

Example (Sample Size Calculation).

Suppose we have $\hat{\sigma}^2$ under SRS, how wide is my 95% CI if

- (a) Sampled 5% of population, i.e. $\frac{n}{N} = 0.05$?
- (b) Sampled 100 people?

Proof.

- (a) We don't know since we need to know sample size n or population size N .

(b) We don't know since we need to know $\frac{n}{N}$ or N .

□

We need information on both n and N !

For the 95% CI for μ , the total CI width is

$$2 \times 1.96 \times \sqrt{\left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}}.$$

So if we want the total width $< c$, this means

$$\begin{aligned} \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n} &< \left(\frac{c}{2 \cdot 1.96}\right)^2 \\ \frac{1}{n} - \frac{1}{N} &< \frac{1}{\hat{\sigma}^2} \cdot \left(\frac{c}{2 \cdot 1.96}\right)^2 \\ n &> \left[\frac{1}{\hat{\sigma}^2} \left(\frac{c}{2 \cdot 1.96}\right)^2 + \frac{1}{N} \right]^{-1}. \end{aligned}$$

Don't forget to round n to integer!

Where to get $\hat{\sigma}^2$?

- Previous survey.
- In the special case of proportions,

$$\hat{\sigma}^2 = \frac{n}{n-1} \hat{p}(1 - \hat{p}) \approx p(1 - p) \leq 0.5(1 - 0.5) = 0.25.$$

This means that we can use 0.25 as a conservative upper bound for $\hat{\sigma}^2$.

Analyzing a SRS in R

The sampling designs in this class can be analyzed using the R package survey.

```
install.packages("survey") # only need to run once
library(survey) # load the package

srs <- svydesign(ids=~1, data = dat, fpc = ~N)
# "ids=~1" tells R its a SRS, no clusters or strata
# "fpc" tells R which column is pop size to use for fpc
svymean(~sleep, srs)
# get estimate and SE for mean of variable "sleep"
```

```
mean(dat$sleep)
sqrt((1 - 1249/48415) * var(dat$sleep)/1249)
# can also manually calculate mean and SE like this
confint(svymean(~sleep, srs))
# get 95% CI for mean of "sleep"
```

Lecture 5, 2025/09/17

1.3 Questionnaire Design

When a survey respondent reads or hears a survey question, the following process usually occurs in their mind:

- Comprehension (interpreting the question)
- Retrieval (recall of information)
- Judgment (summarizing the information)
- Reporting (answering the question in the required format)

A well-designed questionnaire should make these steps as easy and efficient as possible for the respondent. If the process breaks down anywhere, answers collected will likely be unreliable (measurement error).

Comprehension problems:

- Poor or ambiguous wording.
- Too complex.
- Leading questions.
- False assumptions or inferences.

Retrieval problems:

- Lack of retrieval cues.
- Things are too hard to remember.
- Events happened too long ago.

Judgment and reporting problems:

- Response categories are too specific or don't cover all possibilities.
- Difficult to estimate based on question constraints.
- Stigma of giving certain responses.
- Deliberate misreporting.
 - People don't want to answer sensitive questions or reluctant to admit.

Response scale effects:

- People often don't want to pick options on the ends.
- Scale of -5 to $+5$ is interpreted differently than 0 to 10 . Respondents tend to use higher ratings on the -5 to $+5$ scale.

Types of responses:

- Open-ended, with numerical response.
- Closed questions with ordered response scales.
- Closed questions with categorical response options.
 - Should be disjoint, and cover all possibilities.
 - Be specific.
 - Order matters: some respondents stop reading when they see an option that seems to describe them.

Strategies for sensitive questions:

- Given them a reasonable excuse to soften the behavior.
- Make it as anonymous as possible.
- Randomized response.

Question testing:

- Survey questions often need multiple rounds of testing and feedback.